VBF-C1T1R1 design_spec - Virtual Battery Factory deliverable

Design Specification - Next-Generation Pure Electric Sedan Cell
VBF-C1T1R1-DS-001 | Case c1_t1_r1 | Virtual Battery Factory

1 Objective (verbatim task text)
Design a battery for a next-generation pure electric sedan: energy density >= 392.61 Wh/kg, support

2 Acceptance criteria (entry-0 pre-registration, mechanical judgment basis)

Metric	Threshold	Decision layer
energy_density_wh_kg	>= 392.61	stage2 (calc-energy contract formula, electrolyte excluded)
plated (4C fast charge, 45 C ambient)	false (anode potential min >= 0 V)	stage3
T_max_K (every simulated scenario)	<= 333.15 K (= 60 C)	stage3
triggered (overcharge to 4.7 V)	false	stage3

3 System baseline
Chen2020 (anchor-table default - task names no electrode system): NMC811/graphite,
4.2 V upper cut-off; cell area 0.1027 m2 (0.065 x 1.58 m); 1C current = 5 A.
Electrolyte excluded from energy-density contract (no density in the parameter set).

4 Champion design: R4-V3-margin (Chen2020 base + 9 overrides)

Parameter	Base Chen2020	Champion	Rationale
Positive current collector thickness [m]	16e-6	8e-6	ED lever (inactive mass), no electrochem
Negative current collector thickness [m]	12e-6	6e-6	ED lever (inactive mass)
Separator thickness [m]	12e-6	8e-6	ED lever
Separator porosity	0.47	0.55	ion transport to the plating front
Cation transference number	0.2594	0.7	kills concentration polarization at 20 A (plating root
Electrolyte diffusivity [m2.s-1]	Nyman2008 fn	2e-9	plating margin
Electrolyte conductivity [S.m-1]	Nyman2008 fn	3.0	plating margin
Negative particle radius [m]	5.86e-6	3e-6	anode kinetics/transport, plating margin
Total heat transfer coefficient [W.m-2.K-1]	10	100	4C T_max control

Unchanged from base: positive thickness 75.6 um, negative thickness 85.2 um,
positive particle radius 5.22 um, positive porosity 0.335, negative porosity 0.25.

5 Measured performance (SPMe tool outputs; all values from output files)

Metric	Measured	Criterion	Verdict
Energy density (calc-energy contract)	502.70 Wh/kg	>= 392.61	PASS (+110.09)
Volumetric energy density	953.34 Wh/L	-	-
Contract capacity / energy	5.0306 Ah / 17.8975 Wh	-	-
Cell mass	35.603 g	-	-
Midpoint voltage	4.0073 V	-	-
DCR	0.14679 mOhm (baseline 0.34755 mOhm, -58%)	-	-
Power density	797.3 kW/kg	-	-
4C charge at 45 C ambient: T_max	323.98 K (50.83 C)	<= 333.15	PASS
4C charge: anode potential min	+0.0503 V	>= 0	PASS (no plating)
Overcharge to 4.7 V: reached / T_max	4.70 V / 299.45 K	<= 333.15	PASS
Thermal runaway model (run-tr --sim coupling)	triggered=false	false	PASS

6 Design path (R1-R4)
R1 baseline: ED 400.29 pass; 4C plating fail (anode min -0.4385 V).
R2 architecture levers (thin CC/separator, fast-charge kit, N/P up, combo):
ED up to 566.3, plating persists - diagnosed: anode-potential minimum occurs at the
END of the 4C charge (V=4.2 cut-off), anode-side polarization dominates.
R3 electrolyte transport kit (t+ 0.6, D_e 1e-9, kappa 2.0, negative radius 4 um):
plating eliminated (anode min +0.0420 V) but plating-free charge now accepts more
current: 4C T_max 335.77 K overshoot.
R4 cooling h=100 + transport margin (t+ 0.7, D_e 2e-9, kappa 3.0, radius 3 um):
all criteria pass. Cooling-kinetics trade-off: h=150 (R4-V4-coolmax) re-plates the
cell (anode min -0.0006 V - cooler cell slows kinetics).
Backup R4-V1-cool also passes all criteria (ED 499.43, anode min +0.0247 V,
4C T_max 325.08 K).

7 Honesty notes

- Electrolyte mass excluded from ED (calc-energy contract; parameter set lacks density).
- All simulations SPMe (lumped thermal); DFN not run for the champion.
- Chen2020 first-cycle low-capacity artifact: 1C-protocol run shows 1.118 Ah while the calc-energy contract integration gives 5.0306 Ah; ED uses the contract value.
- True compute (DFT/MD) not run: real_compute=false; endorse entry skipped honestly.
- Plating-free margin is +50 mV on anode potential - manufacturing window is documented, not guaranteed by proxy simulation.